

Exact Tetragonal Coordinates for the Genus-4 M_{23} Quotient

Symmetric Principal Parts, Exact Arithmetic Closure, and the Seven-Band Support Law

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Abstract

The genus-4 quotient $X/\mathbb{Q}$ in the regular M_{23} realization carries a degree-23 function t and a tetragonal pencil. This article makes the pencil exact in two coordinates. Let b', c' be the conjugate totally ramified points above $q = t^2 + 23 = 0$. A Galois-symmetric principal-part normalization of $L(\operatorname{div}(dt) - b' - c')$ defines a degree-4 coordinate w satisfying $w(P)t(P) = 1$ at both points. In the integral coordinate $u = qv$, the monic model $G(t, u) = 0$ has $\operatorname{ord}_{P_{\pm}}(q) = 23$, $\operatorname{ord}_{P_{\pm}}(u) = 19$, and $\operatorname{ord}_{P_{\pm}}(G_u) = 418$, forcing the ceiling

$$m_j \geq \left\lceil \frac{397 - 19j}{23} \right\rceil.$$

This proves a q^{18} factor, seven support bands with caps $(4, 10, 15, 21 \mid 4, 10, 15)$, and a 15-slot forbidden odd triangle. A parity-adapted frame yields primitive integer adjoints H_1, H_2 and a primitive 120-term relation $R(T, Z)$ of exact bidegree $(4, 23)$; a weighted-degree bound and 971 exact evaluations prove $R(t, H_1/H_2) = 0$ over $\mathbb{Q}$. An explicit matrix in $\operatorname{PGL}_2(\mathbb{Q})$ carries H_1/H_2 to w , giving the exact symmetric equation; the reductions of that equation match all 22 finite-field tables coefficientwise (21 reconstruction primes and one independent held-out prime). The published plane model of Huang, Jackson, Lee, Poonen, Pries, and Zhang fixes the coordinates, and their gonality-4 statement in Remark 3.4 supplies the optimality input.

1 Introduction

This article gives exact tetragonal closure for a genus-4 quotient $X/\mathbb{Q}$ carrying a degree-23 function t in a regular M_{23} realization. The quotient, function, and published plane model come from Huang–Jackson–Lee–Poonen–Pries–Zhang (HJLPPZ) [1]; the plane model fixes the coordinates used throughout, and Remark 3.4 supplies the optimality input that excludes maps of degree at most 3 over $\mathbb{Q}$. The construction below independently recovers the degree-4 pencil, a symmetric two-point normalization, an exact bidegree- $(4, 23)$ equation, and the geometric support law that makes coefficient reconstruction possible. These exact coordinates complement the positive-dimensional Hurwitz-space approach in [2], which develops braid-orbit reconstruction, admissible-cover boundary geometry, and higher-branch refinements for M_{23} .

The central object is a primitive polynomial

$$R(T, Z) = \sum_{k=0}^{23} \sum_{d=0}^4 c_{k,d} Z^k T^d \in \mathbb{Z}[T, Z]$$

with all 120 coefficient slots nonzero and maximum coefficient size 189 decimal digits. Its arithmetic coordinate z generates $\mathbb{Q}(X)$ together with t and has degree 4. A characteristic-zero principal-part calculation gives an exact constant transformation $z \leftrightarrow w$ in $\operatorname{PGL}_2(\mathbb{Q})$ to the Galois-symmetric

tetragonal coordinate. The transformed symmetric equation has coefficient heights up to 3292 digits. Thus, a 133-digit symmetric-frame CRT modulus lies far below the scale required for direct reconstruction.

The structural theorem is the seven-band law. After $u = (t^2 + 23)v$, canonical adjoints have weighted degree 42. The two totally ramified points force a valuation ceiling on every u^j coefficient. The ceiling reduces a 484-slot weighted lattice to an 86-slot support carrier and isolates a 15-slot odd triangle. The proof uses the ramification data $v(q) = 23$, $v(u) = 19$, $v(G_u) = 418$, and the extra vanishing imposed by $b' + c'$.

Modular calculations support discovery and independent validation, while characteristic-zero arguments prove the final relation and frame transition. A degree bound and 971 exact specializations establish the relation; a six-specialization rank computation supplies an independent uniqueness check. The finite-field comparison uses 21 CRT tables in acceptance order and an independently computed table at $p = 1013$. The exact transition reproduces all 120 symmetric coefficients projectively at every one of these primes, including $p = 100208161$ as a direct validation prime. Section 10 separates proof-bearing computations, corroborating checks, discovery calculations, and bounded diagnostics.

Appendix A establishes global ordinary-height minimality for the HJLPPZ Example 3.7 generator under all rational fractional-linear changes. That calculation remains logically separate from the tetragonal theorem.

2 Model and the tetragonal pencil

Let $X/\mathbb{Q}$ be the genus-4 quotient of [1], and let

$$t : X \longrightarrow \mathbb{P}_{\mathbb{Q}}^1$$

be the published degree-23 function. Put

$$q = t^2 + 23, \quad K = \mathbb{Q}(\sqrt{-23}), \quad s^2 = -23.$$

The two totally ramified points above $q = 0$ are

$$P_+ = b', \quad P_- = c', \quad t(P_{\pm}) = \pm s,$$

and the conjugate sum $D = P_+ + P_-$ descends to $\mathbb{Q}$. Let v denote the published degree-8 coordinate and set $u = qv$. At either point $P_{\pm}$, the local valuations are

$$v_P(q) = 23, \quad v_P(u) = 19, \quad v_P(dt) = 22.$$

The Bezout identity $17 \cdot 19 - 14 \cdot 23 = 1$ gives conjugate uniformizers

$$\pi_+ = \frac{u^{17}}{(t-s)^{14}}, \quad \pi_- = \frac{u^{17}}{(t+s)^{14}}.$$

Fix $K_t = \text{div}(dt)$ and define

$$\mathcal{V} = L(K_t - D).$$

Multiplication by dt identifies $\mathcal{V}$ with $H^0(X, \omega_X(-D))$.

Theorem 2.1 (Optimal tetragonal pencil, after HJLPPZ). *The $\mathbb{Q}$ -vector space $\mathcal{V}$ has dimension 2. Every basis (f_1, f_2) defines a rational function f_1/f_2 of degree 4 on X .*

Proof. Riemann–Roch gives $\dim_{\mathbb{Q}} \mathcal{V} = 2$ by [1, Remark 3.4]. The complete pencil $|K_X - D|$ has degree 4. A positive fixed part would leave a rational map to $\mathbb{P}^1$ of degree at most 3. Degree 1 is impossible because X has genus 4, and degree 2 is impossible because X is non-hyperelliptic; the canonical model of X is the quadric–cubic complete intersection described after [1, Lemma 3.3]. Thus, only degree 3 remains. The complex degree-3 maps are the two rulings of the canonical quadric, and [1, Remark 3.4] shows that neither ruling descends to $\mathbb{Q}$. Hence, the pencil is base-point free over $\mathbb{Q}$, and every basis ratio has degree 4. $\square$

3 Symmetric principal-part normalization

The finite-field construction and the characteristic-zero frame calculation use principal parts of Riemann–Roch functions.

For $f \in L(K_t)$, define

$$\lambda_{0,\pm}(f) = \text{red}_{P_{\pm}}(f\pi_{\pm}^{22}) \in K,$$

where red_P denotes reduction in the residue field. Since $v_{P_{\pm}}(f) \geq -22$, these reductions are defined. Membership in $\mathcal{V}$ is equivalent to

$$\lambda_{0,+}(f) = \lambda_{0,-}(f) = 0.$$

For $f \in \mathcal{V}$, define the first surviving principal-part coefficients

$$\lambda_{1,\pm}(f) = \text{red}_{P_{\pm}}(f\pi_{\pm}^{21}).$$

Conjugation exchanges the two coefficients. Hence,

$$A_1(f) = \frac{\lambda_{1,+}(f) + \lambda_{1,-}(f)}{2}, \quad B_1(f) = \frac{\lambda_{1,+}(f) - \lambda_{1,-}(f)}{2s}$$

are $\mathbb{Q}$ -linear functionals on $\mathcal{V}$.

Lemma 3.1 (Smooth proper model at 31). *Let $R = \mathbb{Z}_{31}$ and $K_{31} = \mathbb{Q}_{31}$. The base change $X_{K_{31}}$ has a smooth proper R -model $\mathcal{X}$ such that t extends to a finite morphism*

$$t_R : \mathcal{X} \longrightarrow \mathbb{P}_R^1.$$

The closures $\mathcal{P}_+, \mathcal{P}_-$ of P_+, P_- are disjoint sections of $\mathcal{X}/R$. Along these sections the functions

$$\pi_+ = \frac{u^{17}}{(t-s)^{14}}, \quad \pi_- = \frac{u^{17}}{(t+s)^{14}}$$

are regular relative parameters.

Proof. Since $15^2 \equiv -23 \pmod{31}$ and $2 \cdot 15 \not\equiv 0 \pmod{31}$, Hensel’s lemma gives $s \in R$ with $s^2 = -23$. In homogeneous coordinates $[T : S]$ on $\mathbb{P}_R^1$, let

$$\mathcal{B} = V(T(T - sS)(T + sS)).$$

This effective Cartier divisor is the disjoint union of the three sections $t = T/S = 0, s, -s$: their pairwise differences are $s, -s, 2s$, all 31-adic units. In particular, $\mathcal{B}$ is regular and finite *étale* over R . Put $L = K_{31}(X)$ and define $\mathcal{X}$ to be the normalization of $\mathbb{P}_R^1$ in L . The scheme $\mathbb{P}_R^1$ is excellent, so the normalization is finite; hence $\mathcal{X}$ is proper over R , and the generic fiber is the smooth projective curve $X_{K_{31}}$.

It remains to prove smoothness. Let η_{31} be the vertical codimension-one point of $\mathbb{P}_R^1$ and $A_0 = \mathcal{O}_{\mathbb{P}_R^1, \eta_{31}}$, a DVR with uniformizer 31 and residue field $\mathbb{F}_{31}(t)$. Here, $q = t^2 + 23$ is a unit, so the change $u = qv$ is invertible. The integral u -equation used below is monic of degree 23 and therefore defines a finite free A_0 -algebra B_0 of rank 23.

Exact factorization gives that the specialization $G(7, U) \in \mathbb{F}_{31}[U]$ is irreducible of degree 23, with derivative relatively prime to $G(7, U)$. If the monic polynomial $G(t, U)$ factored over $\mathbb{F}_{31}(t)$, Gauss's lemma would give a factorization into monic positive-degree polynomials in $\mathbb{F}_{31}[t, U]$, and specialization at $t = 7$ would preserve their U -degrees, contradicting irreducibility. Thus, G is irreducible over $\mathbb{F}_{31}(t)$. Since $31 \nmid 23$, an irreducible polynomial of degree 23 in characteristic 31 cannot be inseparable. Hence, $B_0/31B_0$ is a separable field extension of degree 23. The reduction of G_u is nonzero in that field and therefore a unit; Nakayama makes G_u a unit in B_0 . Consequently, B_0/A_0 is finite *étale*. Since q is invertible in $K_{31}(t)$, one has $K_{31}(t, u) = K_{31}(t, v) = L$; thus the fraction field of B_0 is L . As a finite étale algebra over the DVR A_0 , B_0 is normal, so it is already the integral closure of A_0 in L . Thus, the normalization has no vertical ramification at 31. Independently, [1, Proposition 3.5] gives the arithmetic-monodromy statement at 31.

Let $U = \mathbb{P}_R^1 \setminus \mathcal{B}$. By [1, Proposition 3.5], the characteristic-zero branch locus is exactly $0, \pm s$, and infinity is unramified. The preceding paragraph treats the unique vertical codimension-one point. Hence, the normalization over U is unramified in codimension one. Since U is regular and $\mathcal{X}_U$ is normal, purity of the branch locus gives a finite *étale* morphism $\mathcal{X}_U \rightarrow U$ [4, Tag 0BMB].

Along $\mathcal{B}$, the ramification indices are 2 over $t = 0$ and 23 over $t = \pm s$ (with unramified points also present above $t = 0$). The codimension-one points above $\mathcal{B}$ lie over the generic fiber and therefore have residue characteristic 0; their residue extensions are separable and their ramification is tame. The vertical codimension-one point is already unramified. Thus, the finite cover of $\mathbb{P}_R^1 \setminus \mathcal{B}$ is tamely ramified over $\mathbb{P}_R^1$ in codimension one, exactly the hypothesis of Abhyankar's lemma for a regular divisor. Since 2 and 23 are units in R , that lemma gives, étale locally on $\mathbb{P}_R^1$, the standard tame form

$$A[r]/(r^e - \tau), \quad e \in \{1, 2, 23\},$$

where τ cuts out the relevant branch section [4, Tag 0EYG]. Because that section is étale over R , one may choose τ as a relative parameter, so A is étale over $R[\tau]$; the displayed ring is then étale over $R[r]$. It is therefore smooth over R . Together with the étale locus above U , this proves that $\mathcal{X}/R$ is smooth and proper.

The K_{31} -points $P_{\pm}$ extend uniquely to sections by properness. Their reductions remain distinct because $t(P_+) - t(P_-) = 2s$ is a unit. At either section, the standard tame parameter $r_{\pm}$ satisfies

$$t \mp s = a_{\pm} r_{\pm}^{23}$$

with $a_{\pm}$ a unit. The local ring at the closed point of $\mathcal{P}_{\pm}$ is regular, hence factorial. The characteristic-zero valuation $\text{ord}_{P_{\pm}}(u) = 19$ therefore factors

$$u = b_{\pm} r_{\pm}^{19}$$

with $b_{\pm}$ regular. An exact special-fiber calculation gives order 19 as well, so the reduction of $b_{\pm}$ at $r_{\pm} = 0$ is nonzero; hence, $b_{\pm}$ is a unit in the two-dimensional local ring. Consequently,

$$\pi_{\pm} = \frac{b_{\pm}^{17}}{a_{\pm}^{14}} r_{\pm}^{17 \cdot 19 - 14 \cdot 23} = \frac{b_{\pm}^{17}}{a_{\pm}^{14}} r_{\pm},$$

so $\pi_{\pm}$ is regular and has unit linear term. Thus, it is a relative parameter along $\mathcal{P}_{\pm}$. $\square$

Remark 3.2. *The exceptional $p = 31$ collision in the tetragonal plane equation is a singularity of that nonnormal presentation, not a singularity of $\mathcal{X}$. The normalization above is smooth; the plane-model collision disappears on normalization.*

Proposition 3.3 (Nondegenerate symmetric principal-part frame). *The map $J = (A_1, B_1) : \mathcal{V} \rightarrow \mathbb{Q}^2$ is an isomorphism. Consequently, a unique basis (f_A, f_B) satisfies*

$$A_1(f_A) = 1, \quad B_1(f_A) = 0, \quad A_1(f_B) = 0, \quad B_1(f_B) = 1.$$

Proof. Use the smooth proper model $\mathcal{X}/R$ from Lemma 3.1. Write $\rho : \mathcal{X} \rightarrow \operatorname{Spec} R$ for the structure morphism. Since t_R is finite and $\mathbb{P}_R^1$ is projective over R , the morphism ρ is projective. Put

$$\mathcal{D} = \mathcal{P}_+ + \mathcal{P}_-, \quad \mathcal{L} = \omega_{\mathcal{X}/R}(-\mathcal{D}).$$

The divisor $\mathcal{D}$ is relative effective Cartier, so $\mathcal{L}$ is an invertible, hence R -flat, sheaf. Riemann–Roch gives $h^0(\mathcal{L}_{K_{31}}) = 2$ on the generic fiber. On the special fiber, the $p = 31$ differential computation gives a 4-dimensional canonical space and two independent zeroth principal-part conditions, so $h^0(\mathcal{L}_{\mathbb{F}_{31}}) = 2$ as well. These are the only two fibers of the trait $\operatorname{Spec} R$; hence h^0 is constant. The cohomology-and-base-change theorem implies that $\rho_*\mathcal{L}$ is a free R -module of rank 2 and that base change identifies the generic and special fibers with the corresponding spaces of sections; this is the constant-dimension case of [3, Theorem III.12.11].

Lemma 3.1 shows that $\pi_{\pm}$ are relative parameters, while the opposite factors $t \pm s$, together with 2 and $2s$, are units along the relevant sections. In the tame chart $t \mp s = a_{\pm}r_{\pm}^{23}$, relative differentiation gives

$$dt = r_{\pm}^{22}\varepsilon_{\pm} dr_{\pm}, \quad \varepsilon_{\pm} \in \mathcal{O}_{\mathcal{X}, \mathcal{P}_{\pm}}^{\times},$$

because $23a_{\pm}$ is a unit modulo $r_{\pm}$. Thus, for a relative differential $\omega = f dt \in \mathcal{L}$, the product $f\pi_{\pm}^{21}$ is regular along $\mathcal{P}_{\pm}$ and the restriction of $f\pi_{\pm}^{21}$ to the section lies in R . The trace and skew-trace formulas therefore define an R -linear first-jet map

$$J_R : \rho_*\mathcal{L} \longrightarrow R^2$$

whose generic fiber is the original J and whose formation commutes with reduction. Reduction at $p = 31$ gives

$$\begin{pmatrix} 4 & 28 \\ 25 & 27 \end{pmatrix} \in M_2(\mathbb{F}_{31}), \quad \det = 28 \in \mathbb{F}_{31}^{\times}.$$

Thus, J_R reduces to an isomorphism on the closed fiber. Its determinant is a unit (equivalently, apply Nakayama to the cokernel), so J_R itself is an isomorphism. Passing to the generic fiber identifies $J_R \otimes_R K_{31}$ with $J \otimes_{\mathbb{Q}} K_{31}$; faithful scalar extension then gives the asserted isomorphism $J : \mathcal{V} \rightarrow \mathbb{Q}^2$. The characteristic-zero calculation in Theorem 8.2 supplies an independent exact recovery of the same frame. $\square$

Definition 3.4 (Symmetric tetragonal coordinate). *Set $w = f_A/f_B \in \mathbb{Q}(X)$.*

Proposition 3.5 (Values at the totally ramified points). *The symmetric coordinate satisfies*

$$\boxed{w(P_{\pm})t(P_{\pm}) = 1}.$$

Proof. The normalized principal parts give

$$\lambda_{1,+}(f_A) = \lambda_{1,-}(f_A) = 1, \quad \lambda_{1,+}(f_B) = s, \quad \lambda_{1,-}(f_B) = -s.$$

At each point, the value of the ratio equals the ratio of the first surviving Laurent coefficients. Thus, $w(P_+) = 1/s$ and $w(P_-) = -1/s$, and multiplication by $t(P_\pm) = \pm s$ gives the identity. $\square$

Theorem 3.6 (Exact geometric bidegree). *The functions t and w generate $\mathbb{Q}(X)$. Their image in $\mathbb{P}_t^1 \times \mathbb{P}_w^1$ is an integral curve of bidegree $(4, 23)$.*

Proof. The extension $\mathbb{Q}(X)/\mathbb{Q}(t)$ has prime degree 23, while w has degree 4. Membership $w \in \mathbb{Q}(t)$ would force the degree of w on X to be divisible by 23. Hence, $\mathbb{Q}(t, w)$ is a proper enlargement of $\mathbb{Q}(t)$ inside a prime-degree extension, giving $\mathbb{Q}(t, w) = \mathbb{Q}(X)$. For an integral curve of bidegree (a, b) in $\mathbb{P}^1 \times \mathbb{P}^1$, the first and second projections have degrees b and a ; here these degrees are 23 and 4. $\square$

4 Finite-field realization

The modular pipeline starts from the published degree-8 plane equation of [1]. An integral reparameterization sets

$$q = t^2 + 23, \quad u = qv,$$

where v denotes the published degree-8 coordinate. At a split prime p , write $q = (t - s_+)(t - s_-)$. The conjugate degree-2 place above $q = 0$ splits into two rational places; at either place the local valuations are

$$v_P(q) = 23, \quad v_P(u) = 19.$$

The Bezout identity

$$17 \cdot 19 - 14 \cdot 23 = 1$$

therefore gives a local uniformizer

$$\pi = \frac{u^{17}}{(t - s)^{14}}.$$

The implementation performs four exact steps at each accepted prime:

1. compute a 4-dimensional canonical space;
2. impose the two zeroth principal-part conditions at the conjugate points to recover the 2-dimensional tetragonal pencil;
3. apply the principal-part trace/skew-trace normalization of §3;
4. form the multiplication matrix of w on the degree-23 function field and take the characteristic polynomial of that matrix.

At every accepted prime, the run verifies $[\mathbb{F}_p(X) : \mathbb{F}_p(t)] = 23$ and returns the cleared characteristic-polynomial relation

$$\deg_W H_p = 23, \quad \deg_T H_p = 4.$$

This bidegree verifies generation prime by prime. Indeed, if the normalized coordinate lay in $\mathbb{F}_p(t)$, multiplication by it would be scalar and the characteristic polynomial would be a 23rd power; after primitive denominator clearing, the T -degree of that polynomial would be 0 or a multiple of 23, not 4. Since the ambient extension has prime degree 23, the coordinate therefore generates it, and the characteristic polynomial equals the minimal polynomial. Each successful run also returns a flattened list of $120 = (23 + 1)(4 + 1)$ coefficients.

4.1 Twenty-one-prime CRT checkpoint

The accepted prime list, displayed in acceptance order, is

31, 1000003, 1008001, 1022201, 1028201, 1035301, 1043401,
 1055501, 1074701, 1117111, 1123211, 1126211, 1160611,
 100050001, 100161001, 1183811, 1193911, 1163611,
 100202057, 100207139, 100208161.

Their product has 133 decimal digits, and exact reduction verifies all

$$21 \cdot 120 = 2520$$

coefficient congruences against the CRT vector.

4.2 Independent held-out prime $p = 1013$

The prime 1013 stays outside the CRT reconstruction. The independent finite-field computation reaches the same canonical-space dimension, the same two-dimensional residue kernel, a rank-2 symmetric principal-part matrix, and a relation of bidegree $(23, 4)$ in the program's (W, T) reporting order.

Against the 133-digit training CRT, the independent Python validator records:

$$68 \text{ ordinary rational-reconstruction candidates} \longrightarrow \boxed{1 \text{ held-out survivor}},$$

with survivor index 0 and value 1. Ten five-coefficient rows admit formal projective reconstructions, and zero rows survive the held-out table. Two adjacent ten-coefficient row pairs admit formal projective reconstructions, and zero pairs survive.

The held-out calculation supplies an independent consistency check on the normalized modular relation. It also shows that coefficientwise and small projective rational reconstruction remain below the true characteristic-zero height scale.

5 Height and reconstruction strategy

The 21-prime symmetric-frame CRT has a 133-digit modulus, and direct rational reconstruction in that frame does not survive the independent $p = 1013$ test. The exact calculation explains the obstruction quantitatively: the compact arithmetic-frame equation has maximum coefficient size 189 digits, while the primitive symmetric-frame equation obtained from the exact transition has maximum coefficient size 3292 digits. Thus, adding a small number of primes in the symmetric frame does not address the relevant height scale.

The reconstruction first recovers the two-dimensional adjoint plane, then changes to a parity-adapted arithmetic basis. CRT and lattice calculations only discover exact candidates. Section 8 proves the final relation by characteristic-zero division and exact evaluation.

6 Adjoint space and slot arithmetic

The dense coefficient reconstruction admits a cleaner modular representation. Let $G(t, u) = 0$ be the integral degree-23 plane model, let $G_u = \partial G / \partial u$, and let C_1, C_2 denote the two symmetric normalized canonical functions, so $w = C_1 / C_2$. At the three generic split primes

$$1013, \quad 100207139, \quad 100208161,$$

the replay computations give the same section profile: in the published v -basis each C_i has a common denominator of degree 85 and cleared numerator degree 83, while the quotient w has a common denominator of degree 88. In the integral $u = qv$ basis, $q = t^2 + 23$, the denominator of the u^j coordinate follows the exact staircase

$$D_j = D_0 q^{j - \lceil 4j/23 \rceil}.$$

The exponent is the Newton slope $4/23$ written in the power basis.

More importantly, define

$$h_i = C_i G_u.$$

At all three generic primes, the h_i are polynomials and satisfy

$$h_i = \sum_{j=0}^{21} a_{i,j}(t) u^j, \quad \deg_t a_{i,j} \leq 42 - 2j.$$

Thus, h_i has weighted degree 42 for $\text{wt}(t) = 1$, $\text{wt}(u) = 2$, and

$$\boxed{w = h_1/h_2}.$$

The $p = 1013$ fixed-factor computation gives $\gcd(h_1, h_2) = 1$. The fixed weighted support has $43 + 41 + \dots + 1 = 484$ slots. Both large generic primes contain 469 nonzero slots in each adjoint; at $p = 1013$, h_2 has 469 and h_1 has 468, consistent with one accidental modular zero. The gain is denominator elimination in a fixed weighted polynomial space; support density remains high. The prime 31 preserves polynomiality and the weighted bound but exhibits additional degree cancellations, so it serves as a structural specialization rather than a coefficient-level reconstruction prime.

7 Seven-band support law for the adjoint two-plane

An invertible basis change preserves the two-dimensional adjoint plane. Coefficient recovery therefore reconstructs the plane before choosing an ordered basis. Write $q = t^2 + 23$. The full weighted-degree-42 monomial lattice contains 484 slots. Splitting by parity in t gives

$$253 \text{ even slots}, \quad 231 \text{ odd slots}.$$

The high- u odd triangle contains exactly 15 forbidden slots, with weight multiplicities

$$(33, 1), (35, 2), (37, 3), (39, 4), (41, 5).$$

Hence, the admissible q -pair carrier has dimension

$$253 + (231 - 15) = 469.$$

Proposition 7.1 (Geometric seven-band support law). *Let $s^2 = -23$, so the two totally ramified points b', c' lie over $t = \pm s$, and write*

$$h(t, u) = \sum_{j=0}^{21} a_j(t) u^j, \quad m_j = \text{ord}_q a_j, \quad q = t^2 + 23.$$

For an adjoint differential in $H^0(K_X - b' - c')$, one has

$$m_j \geq \left\lceil \frac{397 - 19j}{23} \right\rceil \quad (0 \leq j \leq 21).$$

Consequently, $h(t, qv) = q^{18}H(t, v)$ with $H \in \mathbb{Q}[t, v]$, $\deg_t H \leq 6$, $\deg_v H \leq 21$, and the q -pair support is contained in the seven bands

	Even channel				Odd channel		
q -level	18	19	20	21	18	19	20
Maximum v -degree	4	10	15	21	4	10	15

Thus, the seven-band carrier contains exactly

$$5 + 11 + 16 + 22 + 5 + 11 + 16 = 86$$

slots. For $j \geq 16$ the odd part vanishes identically, giving the 15-slot forbidden odd triangle with weight multiplicities $(33, 1), (35, 2), (37, 3), (39, 4), (41, 5)$.

Proof. Work at b' over $K = \mathbb{Q}(s)$ and normalize a uniformizer π by $v_{b'}(\pi) = 1$. Total tame ramification of index 23 gives

$$v_{b'}(t - s) = 23, \quad v_{b'}(dt) = 22, \quad v_{b'}(q) = 23,$$

because $t + s$ is a unit at b' . The published coordinate v has polar order 4 at b' and $u = qv$, hence

$$v_{b'}(u) = 23 - 4 = 19.$$

The integral plane equation is monic of degree 23 in u , and the fiber over $t = s$ is a 23rd power. Locally write $G(t, u) = \prod_{i=1}^{23}(u - u_i(t))$. A Puiseux expansion of the chosen branch has

$$u = c_{19}\pi^{19} + O(\pi^{20}), \quad c_{19} \neq 0.$$

The substitutions $\pi \mapsto \zeta^i \pi$ produce the conjugate branches for a primitive 23rd root of unity ζ . Since $\gcd(19, 23) = 1$,

$$v_{b'}(u_1 - u_i) = 19 \quad (i \neq 1),$$

and therefore

$$v_{b'}(G_u) = \sum_{i \neq 1} v_{b'}(u_1 - u_i) = 22 \cdot 19 = 418.$$

Write the adjoint differential as

$$C dt = \frac{h dt}{G_u}.$$

Membership in $H^0(K_X - b' - c')$ gives $v_{b'}(C dt) \geq 1$, so

$$v_{b'}(h) = v_{b'}(C dt) - v_{b'}(dt) + v_{b'}(G_u) \geq 1 - 22 + 418 = 397.$$

Because $a_j(t) \in \mathbb{Q}[t]$, conjugation of s and $-s$ gives $\text{ord}_{t-s} a_j = \text{ord}_{t+s} a_j = m_j$. Hence,

$$v_{b'}(a_j(t)u^j) = 23m_j + 19j.$$

These valuations are pairwise distinct: if $23m + 19j = 23m' + 19j'$ with $0 \leq j, j' \leq 21$, then $23 \mid 19(j' - j)$, so $j = j'$ and then $m = m'$. A nonarchimedean sum of terms with distinct valuations has valuation equal to the minimum. Thus, every nonzero term satisfies

$$23m_j + 19j \geq 397,$$

which is the stated ceiling bound.

Evaluating the ceiling gives

$$m_j \geq \begin{cases} 18 - j, & 0 \leq j \leq 4, \\ 19 - j, & 5 \leq j \leq 10, \\ 20 - j, & 11 \leq j \leq 15, \\ 21 - j, & 16 \leq j \leq 21. \end{cases}$$

Therefore, $j + m_j \geq 18$ and $q^{18} \mid h(t, qv)$. The weighted-degree bound $\deg_t a_j \leq 42 - 2j$ then gives

$$\deg_t(a_j q^{j-18}) \leq 6,$$

so $\deg_t H \leq 6$. The four lower boundaries above yield the even caps 4, 10, 15, 21. For $j \geq 16$, an odd q -digit at level r would require $r \leq 20 - j$, whereas the valuation bound gives $r \geq 21 - j$; hence all such odd digits vanish. The odd caps are therefore 4, 10, 15, and counting the allowed monomials gives the stated 86 slots and the 1, 2, 3, 4, 5 forbidden-triangle staircase. The argument at c' is the Galois conjugate calculation. $\square$

Remark 7.2 (Load-bearing role of the two-point vanishing). *For the full canonical space, the same calculation would give $m_j \geq \lceil (396 - 19j)/23 \rceil$. This differs from the tetragonal bound only at $j = 16$: the canonical estimate permits $m_{16} = 4$, while vanishing at $b' + c'$ forces $m_{16} \geq 5$. Thus, the extra two-point condition acts precisely at the top row of the forbidden odd triangle.*

Corollary 7.3 (Exact support and row-reduced plane). *The exact integer adjoints satisfy the bound of Proposition 7.1, with equality attained. In the canonical lexicographic row-reduced two-plane the pivot slots are 19, 20, each row has 85 nonzero entries, the support union has size 86, the support intersection has size 84, and the common-zero count is 383.*

Proof. Substitution of the two exact adjoints verifies the ceiling bound for all $j = 0, \dots, 21$, with equality attained. Expanding in the weighted u -basis gives all forced odd-triangle zeros. Exact rational row reduction then gives the stated pivots and support counts. $\square$

The distinction between bases matters for interpreting raw nonzero counts. The pre-row-reduction finite-field kernel vectors have 469 nonzero entries among the 484 weighted slots. In the exact arithmetic good frame, H_1 has 463 nonzero slots because six additional coefficients vanish incidentally, while H_2 has 469; both obey the same 15 forced odd-triangle zeros and the same 86-slot q -pair carrier. Their basis dependence excludes them from the support invariants.

After fixing the two lexicographic pivots, only

$$85 + 85 - 2 = 168$$

nonpivot row entries remain. The parity-paired frame used below changes only the basis of this fixed two-plane; it preserves the seven-band carrier.

8 Exact arithmetic-frame coefficient closure

The lexicographic q-pair frame is canonical but arithmetically unfavorable. On the fixed 86-slot support, the parity-paired columns

$$231 = (\text{EVEN}, j = 15, r = 5, \text{wt} = 40), \quad 469 = (\text{ODD}, j = 15, r = 5, \text{wt} = 41)$$

form an invertible 2×2 pivot block at fifteen generic split primes. Re-row-reducing in this frame produces a complete 168-entry rational reconstruction from the fifteen-prime CRT. This modular reconstruction discovers compact characteristic-zero candidates; the theorem below validates the resulting relation independently over $\mathbb{Q}$.

Write the published degree-23 equation in the v -coordinate as $F(t, v) = 0$ and set

$$q = t^2 + 23, \quad F_{\text{int}}(t, v) = q^4 F(t, v), \quad u = qv.$$

For the integral model $G(t, u) = 0$, direct substitution gives

$$G(t, qv) = q^{19} F_{\text{int}}(t, v), \quad G_u(t, qv) = q^{18} (F_{\text{int}})_v(t, v).$$

The q-pair support lies in seven bands, with counts

$$5, 11, 16, 22, 5, 11, 16, \quad 5 + 11 + 16 + 22 + 5 + 11 + 16 = 86.$$

Equivalently, every recovered adjoint representative satisfies

$$h(t, qv) = q^{18} H(t, v),$$

where

$$\begin{aligned} H(t, v) = & P_{18}(v) + qP_{19}(v) + q^2P_{20}(v) + q^3P_{21}(v) \\ & + t(Q_{18}(v) + qQ_{19}(v) + q^2Q_{20}(v)), \end{aligned}$$

with degree bounds

$$(\deg P_{18}, \deg P_{19}, \deg P_{20}, \deg P_{21}) \leq (4, 10, 15, 21), \quad (\deg Q_{18}, \deg Q_{19}, \deg Q_{20}) \leq (4, 10, 15).$$

Thus, the common factor cancels exactly in the differential representation:

$$\frac{h \, dt}{G_u} = \frac{H \, dt}{(F_{\text{int}})_v}.$$

Fix primitive integer polynomials $H_1, H_2 \in \mathbb{Z}[t, v]$, each of v -degree 21 and t -degree 6, and define

$$z = \frac{H_1(t, v)}{H_2(t, v)} \in \mathbb{Q}(X).$$

A finite-field kernel calculation then reconstructs a primitive polynomial

$$R(T, Z) = \sum_{k=0}^{23} \sum_{d=0}^4 c_{k,d} Z^k T^d \in \mathbb{Z}[T, Z].$$

Using 42 primes near 10^9 , the CRT modulus has 379 decimal digits; all 120 normalized coefficients reconstruct, and independent reductions at $p = 1000003$, 1008001 , and 1013 agree. Clearing denominators gives $\gcd(c_{k,d}) = 1$, all 120 coefficients nonzero, and maximum coefficient size 189 decimal digits.

Theorem 8.1 (Exact arithmetic-frame tetragonal equation). *The polynomial R satisfies*

$$R(t, z) = 0 \quad \text{in } \mathbb{Q}(X),$$

with exact bidegree

$$\deg_T R = 4, \quad \deg_Z R = 23.$$

Moreover, $z \notin \mathbb{Q}(t)$, hence $\mathbb{Q}(t, z) = \mathbb{Q}(X)$, and $z : X \rightarrow \mathbb{P}^1$ has degree 4. Consequently, R is the primitive characteristic-zero defining equation of the image of (t, z) , unique up to sign.

Proof. Let

$$R(T, Z) = \sum_{k=0}^{23} A_k(T) Z^k, \quad \deg A_k \leq 4,$$

and homogenize after substituting $z = h_1/h_2$:

$$S(t, u) = \sum_{k=0}^{23} A_k(t) h_1(t, u)^k h_2(t, u)^{23-k}.$$

For the weights $\text{wt}(t) = 1$, $\text{wt}(u) = 2$, the integral equation G is monic of u -degree 23 and has weighted degree at most 46, while each h_i has weighted degree at most 42. Therefore,

$$\text{wt}(S) \leq 4 + 23 \cdot 42 = 970.$$

Division by the monic polynomial G preserves this weighted-degree bound, so every coefficient of the remainder in the basis $1, u, \dots, u^{22}$ has t -degree at most 970. Exact evaluation of the remainder at the 971 distinct integers

$$-485, -484, \dots, 484, 485$$

gives zero at every value. Each remainder coefficient is therefore the zero polynomial, proving $S \equiv 0 \pmod{G}$ and hence $R(t, z) = 0$ in $\mathbb{Q}(X)$.

At $t = 0$, the coefficient vectors of $H_1(0, v)$ and $H_2(0, v)$ have a nonzero 2×2 minor in v -degrees 0 and 1; the resulting integer has 94 digits and is nonzero. Hence, H_1 and H_2 are linearly independent in $\mathbb{Q}(t)[v]$. Since both have v -degree $21 < 23 = \deg_v F$, equality $H_1/H_2 \in \mathbb{Q}(t)$ in the field $\mathbb{Q}(t)[v]/(F)$ would force such polynomial proportionality. Therefore, z lies outside $\mathbb{Q}(t)$. Since $[\mathbb{Q}(X) : \mathbb{Q}(t)] = 23$ is prime, the tower law gives $\mathbb{Q}(t, z) = \mathbb{Q}(X)$. The equation $R(t, z) = 0$ has T -degree 4, so $[\mathbb{Q}(X) : \mathbb{Q}(z)] \leq 4$. The global degree-4 optimality argument used in Theorem 2.1 excludes degrees 1, 2, 3, hence $\deg z = 4$. Finally, $\deg_Z R = 23 = [\mathbb{Q}(X) : \mathbb{Q}(t)]$, so R is the minimal polynomial of z over $\mathbb{Q}(t)$ after primitive normalization. $\square$

Theorem 8.2 (Exact transition to the symmetric frame). *Let w be the two-point normalized coordinate of §3 and let $z = H_1/H_2$ be the arithmetic coordinate above. There is a primitive matrix*

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in M_2(\mathbb{Z}), \quad \det M \neq 0, \quad \boxed{w = \frac{Az + B}{Cz + D}},$$

whose entries have 148, 149, 149, 150 decimal digits. Consequently,

$$R_{\text{sym}}(T, W) = \text{pp} \left((A - CW)^{23} R \left(T, \frac{DW - B}{A - CW} \right) \right)$$

is an explicit primitive defining equation in the geometrically normalized coordinate w , again of bidegree $(4, 23)$; here pp denotes primitive part over $\mathbb{Z}$.

Proof. Write $b_{23} = qc$ in the published integral model. At either point above $q = 0$, with $t = \alpha$, $\alpha^2 = -23$, and $r = t - \alpha$, the local valuations satisfy

$$v_P(u) = 19, \quad v_P(q) = v_P(r) = 23,$$

and the initial equation is $u^{23} + c(\alpha)q^{19}$. In each exact good-frame adjoint, the unique lowest principal-part contribution is the q^9u^{10} term. If

$$L_i(t) = \left(\frac{[v^{10}]H_i}{q} \right) \bmod q,$$

then $G_u \sim 23u^{22}$ and $\pi = u^{17}/r^{14}$ at the chosen branch. The first surviving coefficient is therefore

$$\begin{aligned} \text{red}_P \left(\frac{q^9u^{10}L_i(\alpha)}{G_u} \pi^{21} \right) &= \text{red}_P \left(\frac{L_i(\alpha)}{23} q^9u^{345}r^{-294} \right) \\ &= -\frac{c(\alpha)^{15}}{23} L_i(\alpha) \text{red}_P \left(\frac{q}{r} \right)^{294} \\ &= -\frac{c(\alpha)^{15}(2\alpha)^{294}}{23} L_i(\alpha). \end{aligned}$$

Here, $u^{345} = (u^{23})^{15} = -c(\alpha)^{15}q^{285}$ to leading order and $q/r = t + \alpha$ reduces to 2α . Thus, the local initial form gives the displayed constant directly. Reducing the two first-jet values modulo $q = t^2 + 23$ gives the exact 2×2 good-frame principal-part matrix $J_{\text{good}} \in \text{GL}_2(\mathbb{Q})$. The primitive integral scaling of J_{good}^{-1} is the displayed matrix M , so the unique normalized basis of Proposition 3.3 gives the stated fractional-linear relation. □

Remark 8.3 (Frame height). *The exact arithmetic-frame relation has maximum coefficient size 189 digits, whereas Theorem 8.2 produces a primitive symmetric-frame table with maximum coefficient size 3292 digits. The transition matrix itself has entries of 148–150 digits. Thus, symmetric-frame moduli of 133 or 136 digits lie far below the heights of the reconstructed objects. The frame change reduces arithmetic height while preserving the tetragonal fibration.*

Proposition 8.4 (Six-slice uniqueness). *Let M_6 be the exact 138×120 coefficient matrix obtained from the six specializations $t = 0, 1, 2, 3, 4, 5$ for bidegree- $(4, 23)$ relations in the arithmetic frame. Then*

$$\text{rank}_{\mathbb{Q}}(M_6) = 119, \quad \dim_{\mathbb{Q}} \ker(M_6) = 1.$$

Proof. Reduction of the same exact matrix modulo 1000003 has rank 119, so $\text{rank}_{\mathbb{Q}}(M_6) \geq 119$. Theorem 8.1 supplies a nonzero characteristic-zero relation in the kernel, giving $\text{rank}_{\mathbb{Q}}(M_6) \leq 119$. Equality follows, and the kernel is one-dimensional. The existence theorem uses the independent 971-point remainder identity, so this rank computation supplies a separate uniqueness check. □

9 A general ramified-adjoint ceiling law

The seven-band theorem is an instance of a prime-degree valuation principle.

Proposition 9.1 (Ramified-adjoint ceiling law). *Let $t : X \rightarrow \mathbb{P}^1$ have prime degree p , and let P_+, P_- be a conjugate pair of totally ramified points above the roots of a quadratic polynomial $q(t) \in \mathbb{Q}[t]$. Suppose a function u satisfies*

$$v_P(q) = p, \quad v_P(u) = a, \quad \gcd(a, p) = 1,$$

and the local branches of a monic degree- p equation $G(t, u) = 0$ have initial form $u = c\pi^a + O(\pi^{a+1})$ with $c \neq 0$. Let

$$\eta = \frac{h(t, u) dt}{G_u}, \quad h = \sum_{j=0}^{p-2} a_j(t) u^j,$$

with $a_j(t) \in \mathbb{Q}[t]$, and assume $v_{P_{\pm}}(\eta) \geq \rho$. If $m_j = \text{ord}_q a_j$, then every nonzero coefficient satisfies

$$m_j \geq \left\lceil \frac{(p-1)(a-1) + \rho - aj}{p} \right\rceil.$$

Proof. Total ramification gives $v_P(dt) = p-1$. The $p-1$ differences between the chosen local branch of u and the conjugate branches each have valuation a , because $\gcd(a, p) = 1$; hence $v_P(G_u) = (p-1)a$. Thus,

$$v_P(h) \geq \rho - (p-1) + (p-1)a = (p-1)(a-1) + \rho.$$

The conjugate pair forces the same vanishing order at both roots of q , so $v_P(a_j(t)u^j) = pm_j + aj$. For $0 \leq j \leq p-2$, these term valuations are distinct: equality implies $p \mid a(j-j')$, hence $j = j'$. The valuation of the sum equals the minimum term valuation, and each nonzero term satisfies the displayed ceiling. $\square$

Corollary 9.2 (Support-polygon form). *Assume in addition that $\deg_t a_j \leq N - 2j$, and write uniquely*

$$a_j(t) = \sum_{\ell \geq 0} (e_{j,\ell} + t o_{j,\ell}) q(t)^\ell.$$

Put $C = (p-1)(a-1) + \rho$. A nonzero even slot $e_{j,\ell}$ must satisfy

$$\ell \geq \left\lceil \frac{C - aj}{p} \right\rceil, \quad 2\ell + 2j \leq N,$$

and a nonzero odd slot $o_{j,\ell}$ must satisfy the same lower bound together with $2\ell + 1 + 2j \leq N$. Thus, the adjoint support lies in the finite lattice polygon cut out by the valuation half-plane and the weighted-degree half-plane.

Proof. The ceiling law gives the lower bound on every q -digit. The even monomial $q^\ell u^j$ has weighted degree $2\ell + 2j$, while $tq^\ell u^j$ has weighted degree $2\ell + 1 + 2j$. The asserted inequalities follow. $\square$

For the M_{23} quotient, $(p, a, \rho, N) = (23, 19, 1, 42)$ gives

$$(p-1)(a-1) + \rho = 22 \cdot 18 + 1 = 397,$$

recovering Proposition 7.1. Corollary 9.2 produces the seven bands and the forbidden odd triangle directly from the two bounding half-planes.

10 Computational certificates and logical roles

The mathematical arguments above use computation in four distinct roles. A *proof-bearing* certificate verifies an explicit finite calculation used in the cited result. A *corroborative* certificate checks an already proved statement through an independent representation, specialization, or fresh prime. A *discovery* calculation produces candidates that are later accepted by an independent characteristic-zero proof. A *diagnostic* calculation records a bounded search or explains a reconstruction threshold without entering any theorem. No modular reconstruction alone is used to assert a characteristic-zero identity.

ID	Applies to	Logical role
C1	Lemma 3.1; Proposition 3.3	Proof-bearing local inputs: vertical separability, tame indices, relative parameters, special-fiber dimensions, and the nonzero first-jet determinant.
C2	Section 4; arithmetic-frame candidate construction	Discovery and finite-field validation: 21 CRT tables, the independent $p = 1013$ table, denominator tests, and the fifteen-prime q-pair lift. These data do not prove the characteristic-zero equation.
C3	Proposition 7.1; Corollary 7.3	Proof-bearing finite lattice enumeration and exact verification of the ceiling, weighted degrees, forced odd triangle, pivots, and support counts.
C4	Theorem 8.1; Proposition 8.4	Proof-bearing characteristic-zero identity: the degree-970 remainder bound and 971 exact zeros. The six-slice rank computation is an independent uniqueness check.
C5	Proposition 3.5; Theorem 8.2	Proof-bearing exact first-jet calculation and frame matrix. Coefficientwise reductions and the recovered $p = 1013$ inverse map are corroborative.
C6	Proposition 3.5; Theorem 3.6	Corroborative only: all 22 finite-field tables have bidegree $(4, 23)$ and the prescribed ramified fibers; the clean tables give the recorded $T = 0$ factorization and quartic-discriminant profile.
C7	Theorems 8.1 and 8.2	Adversarial corroboration only: two unused split primes give 12/12 end-to-end checks, and three out-of-window values at three fresh primes give 9/9 zero remainders.
C8	Proposition A.1; Theorem A.2; Lemma A.3; Theorem A.4	Proof-bearing exact Rouché, interval, Sturm, determinant/content, and finite enumeration calculations for the full rational fractional-linear orbit.
C9	Height-search context for Appendix A	Diagnostic only: a bounded search over 37,296 matrices and independent modular factorization checks. No global theorem depends on this bounded search.

Archive locations.

- C1 `certificates/good_reduction_p31/`; `certificates/publication/p31_checks/`; `certificates/publication/regression_p31/`.
- C2 `certificates/current_21prime/`; `certificates/heldout_p1013/`; `certificates/common_denominator_133/`; `certificates/qpair_goodframe_v93/`.
- C3 `certificates/publication/band_law/`; `certificates/publication/exact_closure/`.
- C4 `certificates/exact_tetragonal_v93/`; `certificates/publication/exact_tetragonal/`.
- C5 `certificates/publication/frame_transition/`; `certificates/frame_transition_v95/`; `certificates/publication/exact_tetragonal/verify_compactified_fibers_parallel.py`.

- C6** certificates/publication/ramification/.
- C7** certificates/publication/virgin_prime_spotcheck/; certificates/publication/out_of_window_spotcheck/.
- C8** certificates/global_height_minimality/; certificates/global_fractional_linear_height/.
- C9** certificates/lower_height_specialization/.

The principal corroborating outputs are as follows. The ramification audit verifies the bidegree and the two totally ramified fibers at all 22 finite-field primes; it records the additional small-prime collision at $p = 31$ separately. The exact frame transition matches all 120 coefficients projectively at all 22 primes. At $p = 1013$, the inverse map is

$$z = \frac{546w + 938}{283w + 1},$$

with projective ratio 732, and the compactified degree-23 fiber comparison agrees at all 1013 base values. These checks corroborate, rather than replace, the characteristic-zero proofs.

11 Reproducibility and archive integrity

The accompanying archive contains the article source, exact adjoints, arithmetic and symmetric coefficient tables, finite-field tables, characteristic-zero validators, and execution logs. The principal commands are

```
./RUN_FAST_CERTIFICATES.sh
./RUN_EXACT_TETRAGONAL.sh
./RUN_GLOBAL_HEIGHT_MINIMALITY.sh
./RUN_BOUNDED_HEIGHT_SEARCH.sh
./RUN_ALL_CERTIFICATES.sh
./VERIFY_CHECKSUMS.sh
```

The first three runners execute the structural, exact-identity, frame-transition, and global-height proofs. The fast runner uses isolated worker processes and accepts an override through `M23_FAST_WORKERS`. The exact runner accepts independent overrides through `M23_IDENTITY_WORKERS` and `M23_FIBER_WORKERS`. The bounded-height runner is diagnostic, as recorded in C9, and the all-suite runner executes the four suites in sequence. The checksum command verifies every payload file and exact manifest membership. `CERTIFICATE_INDEX.md` gives the exact inputs, expected success markers, and output files for each certificate family.

12 Conclusion

The tetragonal pencil on the HJLPPZ genus-4 quotient has two exact arithmetic descriptions. The symmetric principal-part coordinate w satisfies $w(P)t(P) = 1$ at the conjugate totally ramified points. The parity-adapted coordinate z admits a compact primitive 120-term relation $R(T, Z)$ of bidegree $(4, 23)$, proved directly in characteristic zero by the degree-970 remainder bound and 971 exact evaluations. The prime-degree field argument gives $\mathbb{Q}(t, z) = \mathbb{Q}(X)$ and $\deg z = 4$.

The support structure follows from geometry. The weighted adjoint lattice has 484 slots, the valuation ceiling forces the q^{18} factor and a 15-slot odd triangle, and the seven bands contain 86 slots. The exact adjoints attain the ceiling, so the bound is sharp. Proposition 9.1 and Corollary 9.2 identify the same mechanism and support polygon for prime-degree covers with coprime local exponents.

The exact matrix in $\mathrm{PGL}_2(\mathbb{Q})$ identifies z with w . Its reductions reproduce all 120 coefficients at all 22 finite-field primes, while the independent $p = 1013$ compactified-fiber comparison agrees at all 1013 base values. The arithmetic-frame height of 189 digits and symmetric-frame height of 3292 digits quantify the reconstruction threshold and complete coefficient closure in both frames.

A Global fractional-linear height minimality for HJLPPZ Example 3.7

Huang et al. obtain degree-23 specializations and then use the PARI/GP routines `polredabs` and `polredbest` to seek smaller defining equations [1, Example 3.7]; see also [5, 7]. Let f be their Example 3.7 polynomial. The invertible generator change

$$\beta = \alpha - 1, \quad g(x) = f(x + 1),$$

produces the same degree-23 field and the same splitting field. The resulting polynomial is

$$\begin{aligned} g(x) = & x^{23} + 23x^{22} + 299x^{21} + 2139x^{20} + 8234x^{19} - 8510x^{18} \\ & - 280094x^{17} - 1592842x^{16} - 3657782x^{15} + 3526406x^{14} \\ & + 44362906x^{13} + 63064206x^{12} - 221821982x^{11} - 727073654x^{10} \\ & + 229352458x^9 + 1938799398x^8 - 1207282236x^7 - 435568756x^6 \\ & + 1165914252x^5 + 727883760x^4 - 482357150x^3 - 983586582x^2 \\ & + 60455178x + 242325562. \end{aligned}$$

For ordinary height $H(h) = \max_i |a_i|$,

$$H(f) = 67,357,061,907, \quad H(g) = 1,938,799,398 =: B,$$

a reduction by the factor

$$\frac{H(f)}{H(g)} = 34.741635455676 \dots$$

Proposition A.1 (Global translate minimum). *Among all integral translates $f(x + k)$ with $k \in \mathbb{Z}$, the polynomial $g(x) = f(x + 1)$ uniquely minimizes ordinary coefficient height.*

Proof. The coefficient of x^{20} in $f(x + k)$ is

$$C_3(k) = 1771k^3 + 966k - 598.$$

Its derivative $C'_3(k) = 5313k^2 + 966$ is positive on $\mathbb{R}$. Exact evaluation gives $C_3(104) > B$ and $C_3(-104) < -B$; hence, $|k| \geq 104$ forces $H(f(x + k)) > B$. An exact sweep of the 209 integers $-104 \leq k \leq 104$ gives minimum B uniquely at $k = 1$. $\square$

For the unimodular statement, homogenize

$$\Phi(X, Y) = Y^{23}f(X/Y), \quad \Gamma(X, Y) = Y^{23}g(X/Y) = \Phi(X + Y, Y),$$

and write H_∞ for the maximum absolute coefficient of a binary form.

Theorem A.2 (Global unimodular height minimum). *For every $M \in \mathrm{GL}_2(\mathbb{Z})$ with $|\det M| = 1$,*

$$H_\infty(\Phi \cdot M) \geq B.$$

Modulo multiplication by -1 , equality consists exactly of the four forms generated from Γ by $X \mapsto -X$ and $X \leftrightarrow Y$. In particular, g is globally height-minimal under all integral unimodular fractional-linear generator changes.

Proof. The proof uses the maximum-norm reduction bound of Hutz–Stoll [6, Theorem 4.7 and Remark 4.9], followed by exact finite enumeration. Choose 23 pairwise disjoint rational Rouché disks of radius 10^{-10} around rational approximations to the roots of g . Exact Taylor inequalities prove that each disk contains one root. Since the 23 disks are disjoint and account for all roots of the degree-23 form, Γ is squarefree, as required by the Hutz–Stoll bound. The same disks imply

$$|\alpha_i| < 9, \quad |\alpha_i - \alpha_j| > \frac{1}{10} \quad (i \neq j).$$

For the Julia covariant point $z(\Gamma) = t + ui$, put

$$D_i = (t - \Re \alpha_i)^2 + (\Im \alpha_i)^2 + u^2.$$

The two components of the gradient of $\log R(\Gamma, t + ui)$ are

$$\sum_i \frac{2(t - \Re \alpha_i)}{D_i}, \quad \sum_i \frac{2u}{D_i} - \frac{23}{u}.$$

Exact rational interval arithmetic on the four sides of the rectangle $[-1, 0] \times [2, 3]$ gives the strict sign pattern inward on opposite sides. Poincaré–Miranda, together with uniqueness of the Julia covariant point, therefore gives

$$-1 < t < 0, \quad 2 < u < 3.$$

The root separation yields a coarse invariant lower bound without numerical approximation. For every point of the upper half-space, at most one root lies within $1/20$ of the horizontal coordinate of the point. Thus, $R(\Gamma, z) \geq u^{23}$ and $R(\Gamma, z) \geq 400^{-22}u^{-21}$; the maximum of these two lower bounds is minimized at $u = 1/20$. Hence,

$$\theta(\Gamma) > 20^{-23}.$$

After carrying $z(\Gamma)$ to i , the corresponding unit-sphere root directions satisfy

$$\eta = 1 - \max_{i \neq j} \sqrt{\frac{1 + \langle \phi_i, \phi_j \rangle}{2}} > \frac{1}{594050}.$$

Indeed, the half-angle sine satisfies

$$\sin^2 \frac{\angle(\phi_i, \phi_j)}{2} = \frac{|\alpha_i - \alpha_j|^2 u^2}{(u^2 + |\alpha_i - t|^2)(u^2 + |\alpha_j - t|^2)} > \frac{1}{297025},$$

and $1 - \cos(\vartheta/2) \geq \frac{1}{2} \sin^2(\vartheta/2)$. Proposition 4.3 of [6] therefore permits

$$\varepsilon(\Gamma) > \frac{1}{2 \cdot 594050^{22}}.$$

With $n = 23$ and $H_\infty(\Gamma) = B$, Remark 4.9 of [6] combines Theorem 4.7 with $\|F\| \leq (n+1)H_\infty(F)^2 = 24H_\infty(F)^2$. Consequently, every $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ with $H_\infty(\Gamma \cdot \gamma) \leq B$ satisfies

$$\cosh d(\gamma^{-1}z(\Gamma), i) < 600,000,000,$$

because the exact integer inequality

$$(600,000,000)^{21} > 2^{23} \cdot 24B^2 \cdot 594050^{22} \cdot 20^{23}$$

holds. If $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then

$$\cosh d(\gamma^{-1}z, i) = \frac{|dz - b|^2 + |a - cz|^2}{2u} \geq c^2 + d^2,$$

since $u > 2$. Hence, $|c|, |d| \leq 24494$.

Finally, put $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Since $\Gamma = \Phi \cdot T$, every determinant-1 transform is $\Phi \cdot (T\gamma)$, and $T\gamma$ has the same bottom row as γ . Because a unimodular integral substitution preserves coefficient content, any transform of height at most B must have both column values bounded by B . Exact Sturm isolation of the seven real roots and six real critical points of f , followed by exact integer evaluation of the resulting Thue columns through denominator 24494, leaves, up to simultaneous sign, only

$$(1, 0), \quad (0, 1), \quad (1, 1), \quad (2, 1).$$

All 40 signed column pairs of determinant ± 1 are then substituted exactly. None has height below B ; the equality set gives precisely the four forms stated above. A determinant- -1 substitution differs from a determinant-1 substitution by a coordinate reflection, which preserves height. $\square$

The rational root centers used to locate the disks are discovery inputs only; the Rouché inequalities validate every retained disk exactly.

Every rational fractional-linear change has a primitive integral matrix representative, unique up to sign, and the primitive part of the transformed binary form is the corresponding primitive degree-23 defining polynomial.

Lemma A.3 (Determinant cutoff). *Let $M \in M_2(\mathbb{Z})$ be primitive, meaning that the gcd of the four entries of M is 1, with $D = |\det M| > 0$. Let C be the coefficient content of $\Phi \cdot M$, and put*

$$H_M = \text{pp}(\Phi \cdot M).$$

Here pp denotes primitive part. If $H_\infty(H_M) \leq B$, then $D \in \{1, 2\}$.

Proof. First, control the local content. If $p \nmid D$, reduction of M modulo p is invertible, so the primitive form Φ cannot become the zero form; hence, $p \nmid C$. Suppose $p^e \parallel D$. Smith reduction over $\mathbb{Z}_p$ gives

$$M = U \begin{pmatrix} 1 & 0 \\ 0 & p^e \end{pmatrix} V, \quad U, V \in \text{GL}_2(\mathbb{Z}_p).$$

The right factor preserves coefficient content. Write $\Phi \cdot U = \sum_{j=0}^{23} A_j X^{23-j} Y^j$. If the first column of U reduces to a projective root of multiplicity m of Φ modulo p , then $A_0, \dots, A_{m-1}$ are divisible by p and A_m is a p -adic unit. The diagonal substitution multiplies the j th coefficient by p^{ej} , so the coefficient $A_m p^{em}$ shows directly that $v_p(C) \leq me$.

For $p \neq 2, 23$, one has $m \leq 2$. Indeed, a projective root of multiplicity at least 3 would be a common root of f, f', f'' modulo p , while exact resultants give

$$\gcd(|\text{Disc}(f)|, |\text{Res}(f, f'')|) = 2^{44} 23^{23}.$$

At $p = 2$, direct reduction gives $f(r) \equiv 2 \pmod{4}$ for each residue class $r \pmod{4}$. Thus, a finite projective point contributes valuation exactly 1 when it reduces to a root, while the point at infinity is not a root because f is monic; hence, $v_2(C) \leq 1$. At $p = 23$,

$$f(x) \equiv x^{23} - 1 = (x - 1)^{23} \pmod{23},$$

and exact evaluation gives $f(1) \equiv 184 = 23 \cdot 8 \pmod{23^2}$ and $f'(1) \equiv 0 \pmod{23}$. For $r = 1 + 23a$, Taylor expansion gives $f(r) \equiv f(1) \pmod{23^2}$ because $23 \mid f'(1)$; hence, every 23-adic point in the unique root class has value of exact valuation 1, and $v_{23}(C) \leq 1$.

Put

$$q = \frac{D^{23}}{C^2}.$$

The preceding local estimates show that every p -adic valuation of q is nonnegative, so q is a positive integer. Binary discriminants transform by

$$|\text{Disc}(H_M)| = |\text{Disc}(\Phi)|q^{22}.$$

Since f is irreducible of degree 23, it has no rational projective root; hence, $h(x) = H_M(x, 1)$ has degree 23 and nonzero integral leading coefficient. If $H_\infty(H_M) \leq B$, Hadamard's inequality on the Sylvester matrix of h and h' gives

$$|\text{Disc}(h)|^2 \leq (24B^2)^{22} \left(\frac{23 \cdot 24 \cdot 47}{6} B^2 \right)^{23}.$$

Exact integer extraction then yields

$$q \leq 76,032,571.$$

For a prime $p \neq 2, 23$ dividing D , the content estimate gives $v_p(q) \geq 19e$, but $3^{19} > 76,032,571$. Thus, no such prime occurs. If $23 \mid D$, then $v_{23}(q) \geq 21$ and 23^{21} exceeds the same bound. Finally, if $4 \mid D$, then $v_2(q) \geq 44$ and 2^{44} also exceeds the bound. Hence, $D \in \{1, 2\}$. $\square$

Theorem A.4 (Global rational fractional-linear height minimum). *For every primitive integral matrix M with $\det M \neq 0$,*

$$H_\infty(\text{pp}(\Phi \cdot M)) \geq B.$$

Equivalently, g has minimum ordinary coefficient height over the full $\text{PGL}_2(\mathbb{Q})$ fractional-linear orbit of the generator of f . Modulo multiplication by -1 , equality consists exactly of the four forms generated from Γ by $X \mapsto -X$ and $X \leftrightarrow Y$.

Proof. Lemma A.3 leaves $D = 1$ and $D = 2$. Theorem A.2 handles $D = 1$.

Assume $D = 2$. With $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, write $\Gamma = \Phi \cdot T$ and $N = T^{-1}M$; the bottom row is unchanged. A coordinate reflection reduces determinant -2 to determinant 2 without changing coefficient height, so

$$\gamma = \frac{1}{\sqrt{2}}N \in \text{SL}_2(\mathbb{R}).$$

The local content bound above gives $C \leq 2$. Therefore, if $H_\infty(H_M) \leq B$,

$$H_\infty(\Gamma \cdot \gamma) = 2^{-23/2}CH_\infty(H_M) < B.$$

The Hutz–Stoll bound used in Theorem A.2 applies to every element of $\text{SL}_2(\mathbb{R})$ [6, Theorem 4.7], so

$$\cosh d(\gamma^{-1}z(\Gamma), i) < 600,000,000.$$

Since the Julia point above has imaginary part $u > 2$, the bottom row (c, d) of M satisfies

$$c^2 + d^2 < 1,200,000,000, \quad |c|, |d| \leq 34641.$$

Moreover, every raw leading or trailing coefficient of $\Phi \cdot M$ has absolute value at most $2B$. Exact Sturm isolation of the seven real roots and six real critical points of f , together with the root neighborhoods constructed in Theorem A.2, reduces every possible column, with the sign normalized by a nonnegative lower entry, to

$$(-2, 0), (-1, 0), (0, 1), (1, 0), (1, 1), (2, 0), (2, 1).$$

Allowing independent signs on the two columns, exactly 32 primitive ordered column pairs of determinant ± 2 arise from these classes. Exact substitution and primitive-part normalization give the minimum height

$$372,261,710,848 > B.$$

Thus, determinant 2 produces neither an improvement nor an equality case. The equality classification is therefore exactly the unimodular classification in Theorem A.2. $\square$

Theorem A.4 closes the full rational fractional-linear orbit. Global minimum height among all defining polynomials arising from arbitrary primitive field generators remains outside the present claim.

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